

Tech & Teach

Digital Chip Design Program

RTL Design

Task #05

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8/7/2026

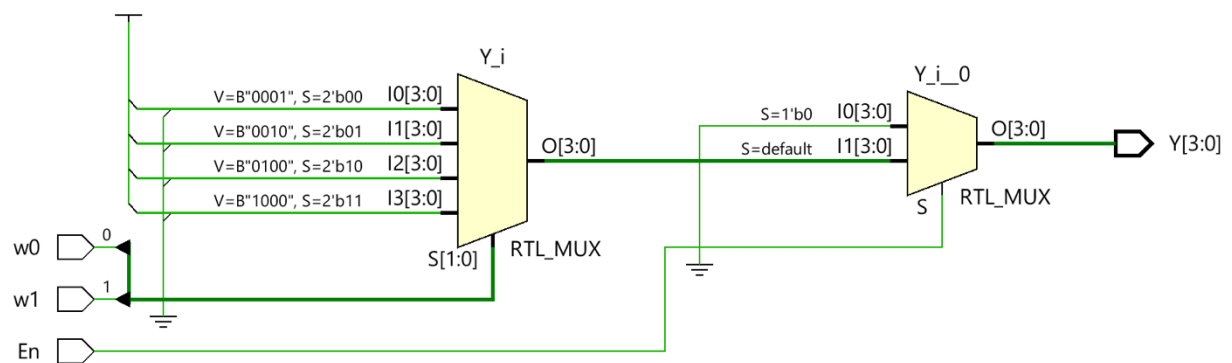
Combinational Datapath Design Part 1

Version 1.0

Task #1.1

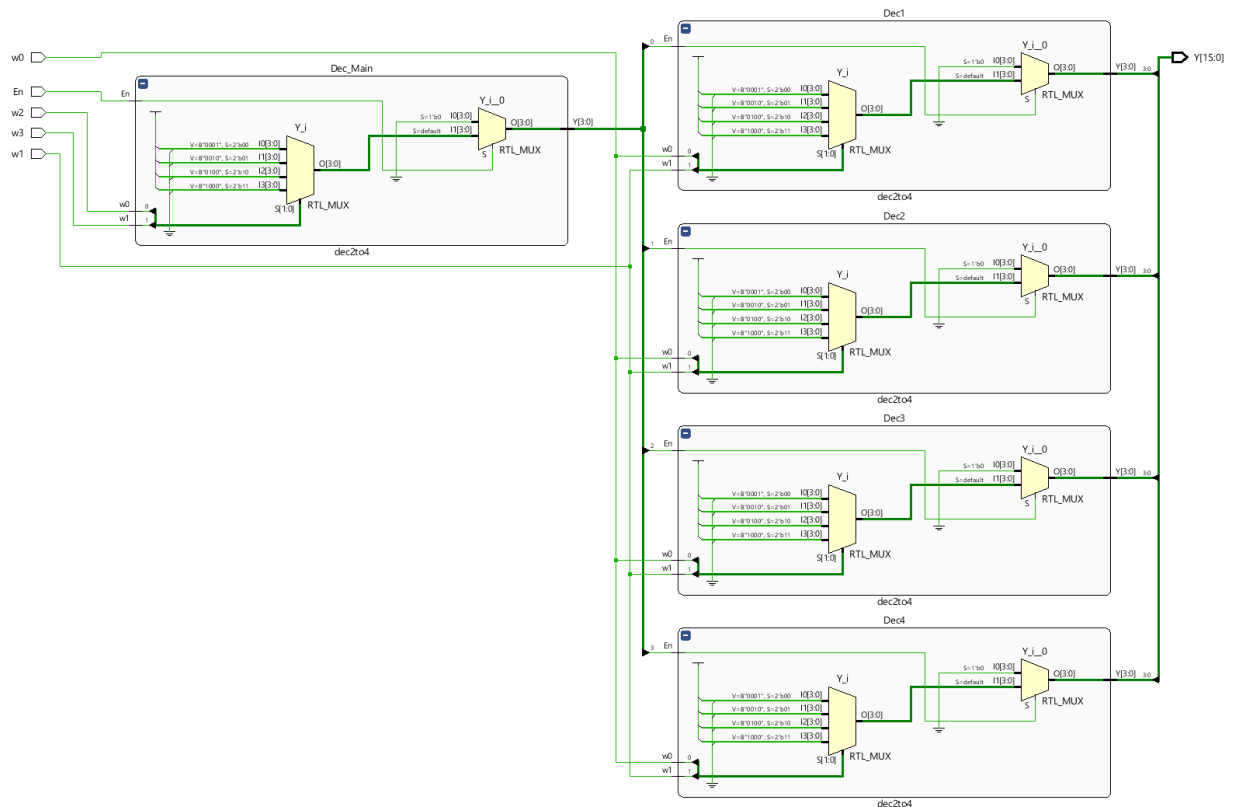
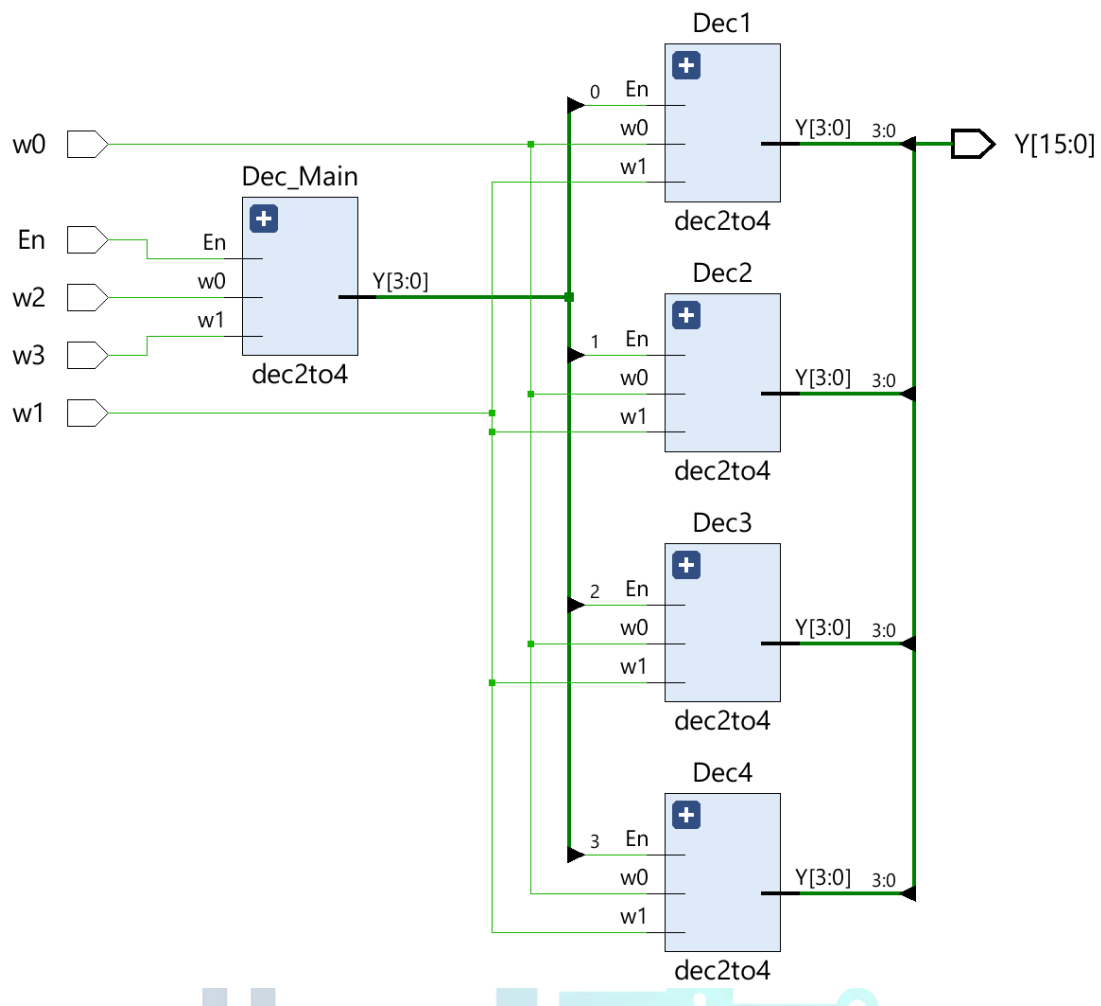
1.1.1

```
1 module dec2to4 (  
2     input logic w1, w0,  
3     input logic En,  
4     output logic [3:0] Y  
5 );  
6  
7     always_comb begin  
8         if (!En) begin  
9             Y = 4'b0000;  
10        end else begin  
11            case ({w1, w0})  
12                2'b00: Y = 4'b0001;  
13                2'b01: Y = 4'b0010;  
14                2'b10: Y = 4'b0100;  
15                2'b11: Y = 4'b1000;  
16                default: Y = 4'b0000;  
17            endcase  
18        end  
19    end  
20  
21 endmodule
```



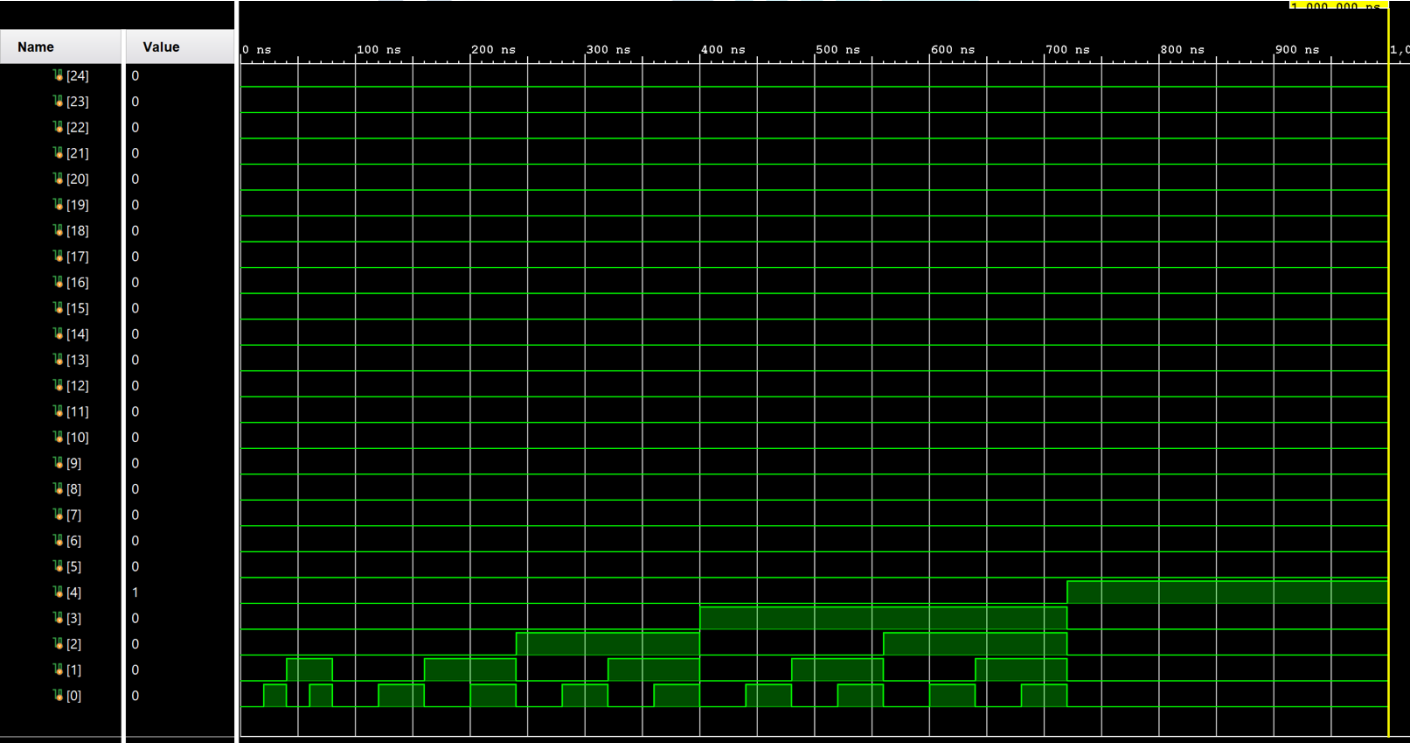
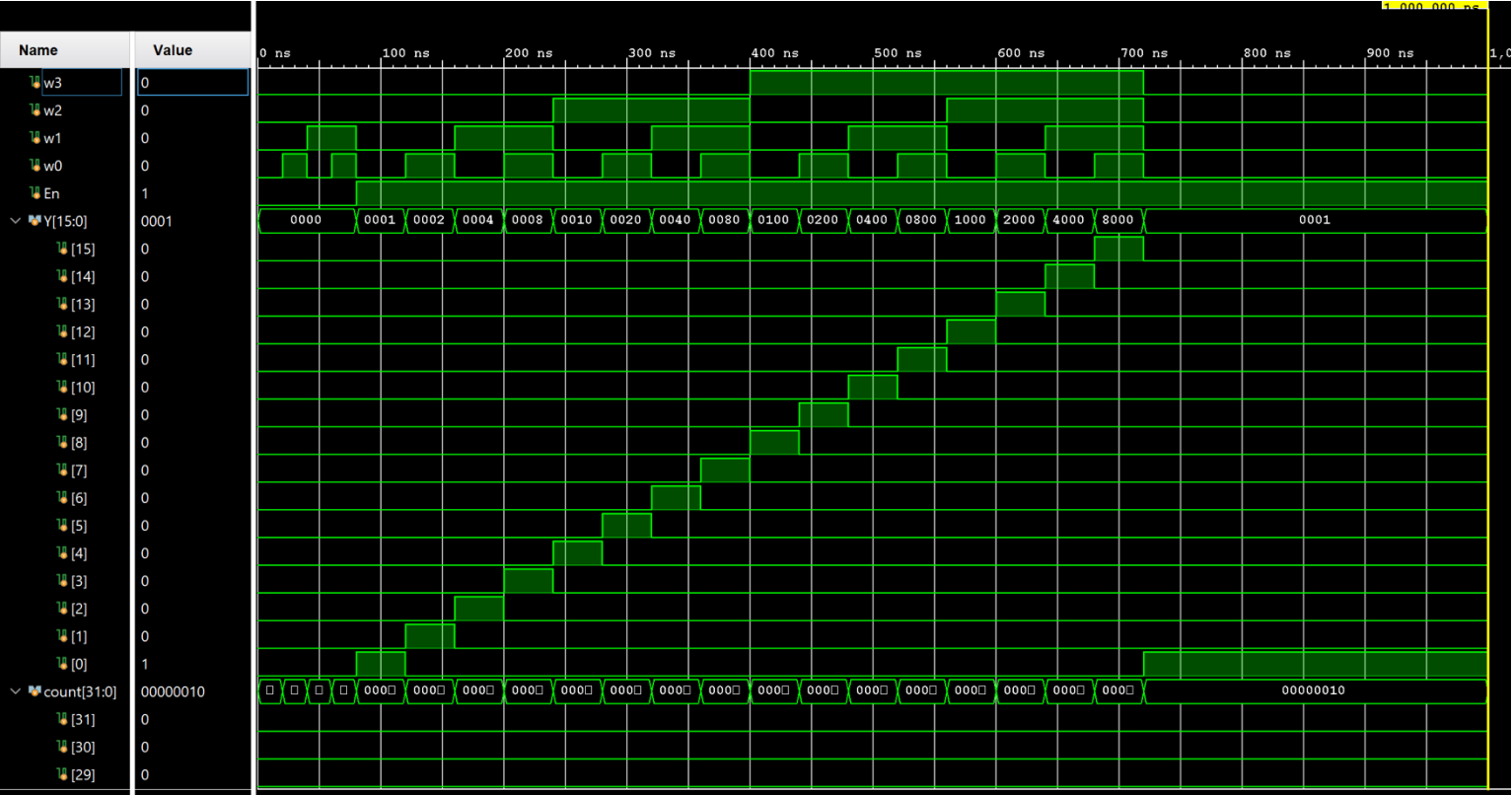
1.1.2

```
1 module dec4to16 (  
2     input  logic w3, w2, w1, w0,  
3     input  logic En,  
4     output logic [15:0] Y  
5 );  
6  
7     logic [3:0] sub_en;  
8  
9     dec2to4 Dec_Main (  
10         .w1(w3),  
11         .w0(w2),  
12         .En(En),  
13         .Y(sub_en)  
14     );  
15  
16     dec2to4 Dec1 (  
17         .w1(w1),  
18         .w0(w0),  
19         .En(sub_en[0]),  
20         .Y(Y[3:0])  
21     );  
22  
23     dec2to4 Dec2 (  
24         .w1(w1),  
25         .w0(w0),  
26         .En(sub_en[1]),  
27         .Y(Y[7:4])  
28     );  
29  
30     dec2to4 Dec3 (  
31         .w1(w1),  
32         .w0(w0),  
33         .En(sub_en[2]),  
34         .Y(Y[11:8])  
35     );  
36  
37     dec2to4 Dec4 (  
38         .w1(w1),  
39         .w0(w0),  
40         .En(sub_en[3]),  
41         .Y(Y[15:12])  
42     );  
43  
44 endmodule
```



1.1.3

```
1  `timescale 1ns/1ps
2  module dec4to16_tb;
3      logic w3, w2, w1, w0;
4      logic En;
5      logic [15:0] Y;
6      logic [3:0] count;
7      assign w3 = count[3];
8      assign w2 = count[2];
9      assign w1 = count[1];
10     assign w0 = count[0];
11
12     dec4to16 dut (
13         .w3(w3),
14         .w2(w2),
15         .w1(w1),
16         .w0(w0),
17         .En(En),
18         .Y(Y)
19     );
20     initial begin
21         $monitor("Time=%0dns | En=%b | W=%b%b%b%b | Y=%16b", $time, En, w3, w2, w1, w0, Y);
22
23         En = 0;
24         count = 4'b0000;
25         repeat(4) begin
26             #10;
27             count = count + 1;
28         end
29         En = 1;
30         count = 4'b0000;
31         for (int i = 0; i < 16; i++) begin
32             #10;
33             count = count + 1;
34         end
35
36         $finish;
37     end
38 endmodule
```



Task #1.2

1.2.1

```
) module half_adder(  
    input logic A, B,  
    output logic S, Cout  
);  
    assign S = A ^ B;  
    assign Cout = A & B;  
) endmodule
```

1.2.2

```
) module full_adder(  
    input logic A, B, Cin,  
    output logic S, Cout  
);  
  
    logic sum1, carry1, carry2;  
  
    half_adder HA1(  
        .A(A),  
        .B(B),  
        .S(sum1),  
        .Cout(carry1)  
    );  
  
    half_adder HA2(  
        .A(Cin),  
        .B(sum1),  
        .S(S),  
        .Cout(carry2)  
    );  
  
    assign Cout = carry1 | carry2;  
) endmodule
```

1.2.3

```
) module ripple_adder(  
    input logic [3:0] A, B,  
    input logic Cin,  
    output logic [3:0] S,  
    output logic Cout  
);  
  
    logic c1, c2, c3;  
  
    full_adder FA0(.A(A[0]), .B(B[0]), .Cin(Cin), .S(S[0]), .Cout(c1));  
    full_adder FA1(.A(A[1]), .B(B[1]), .Cin(c1), .S(S[1]), .Cout(c2));  
    full_adder FA2(.A(A[2]), .B(B[2]), .Cin(c2), .S(S[2]), .Cout(c3));  
    full_adder FA3(.A(A[3]), .B(B[3]), .Cin(c3), .S(S[3]), .Cout(Cout));  
) endmodule
```

1.2.4

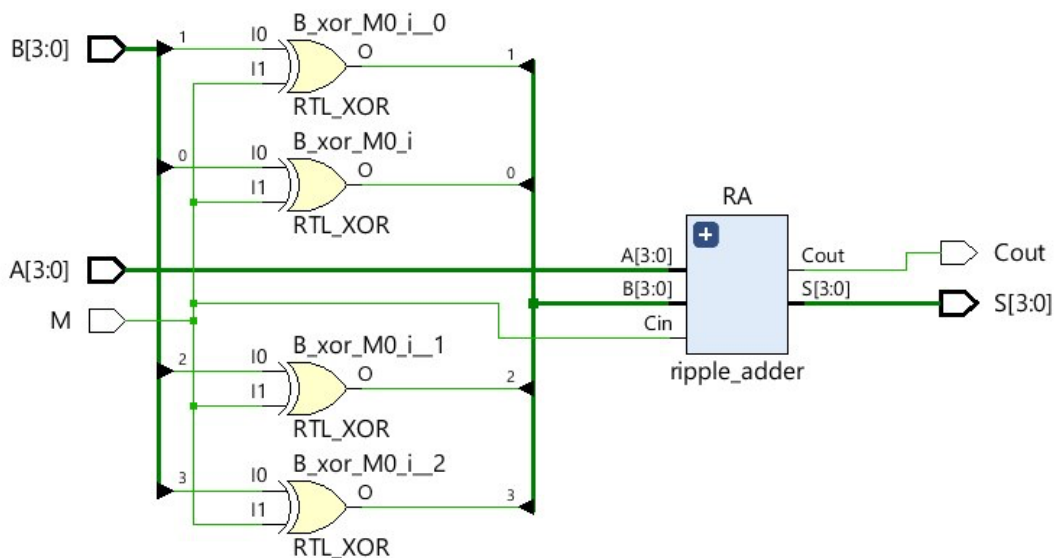
```
) module adder_subtractor(  
    input logic [3:0] A, B,  
    input logic M,  
    output logic [3:0] S,  
    output logic Cout  
);  
  
    logic [3:0] B_xor_M;  
  
    assign B_xor_M[0] = B[0] ^ M;  
    assign B_xor_M[1] = B[1] ^ M;  
    assign B_xor_M[2] = B[2] ^ M;  
    assign B_xor_M[3] = B[3] ^ M;  
  
    ripple_adder RA(  
        .A(A),  
        .B(B_xor_M),  
        .Cin(M),  
        .S(S),  
        .Cout(Cout)  
    );  
) endmodule
```

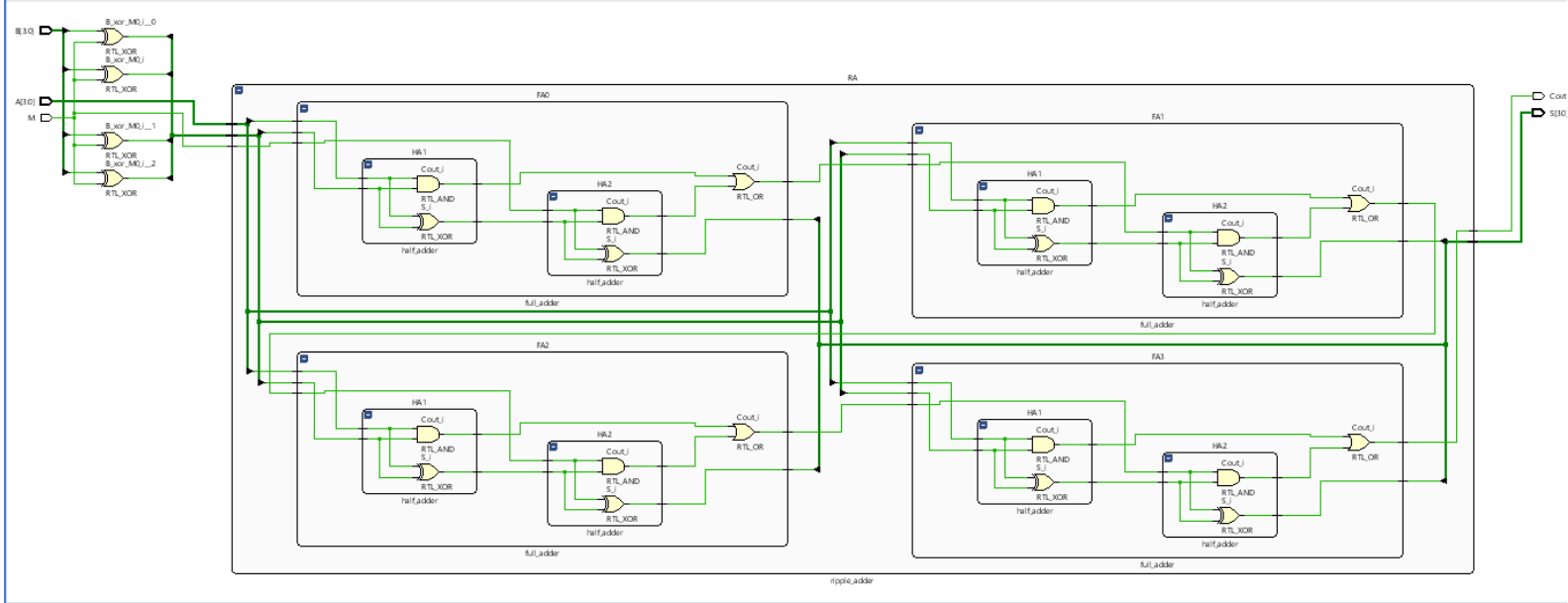


```

21 module tb_adder_subtractor();
22     logic [3:0] A;
23     logic [3:0] B;
24     logic M;
25     logic [3:0] S;
26     logic Cout;
27     adder_subtractor uut (
28         .A(A),
29         .B(B),
30         .M(M),
31         .S(S),
32         .Cout(Cout) );
33     initial begin
34         M = 0;
35         A = 4'b0011; B = 4'b0010; // 3 + 2 = 5
36         #10;
37
38         A = 4'b0111; B = 4'b0001; // 7 + 1 = 8
39         #10;
40
41         M = 1;
42         A = 4'b0111; B = 4'b0010; // 7 - 2 = 5
43         #10;
44
45         A = 4'b0100; B = 4'b0001; // 4 - 1 = 3
46         #10;
47         $finish;
48     end
49 endmodule

```





1.2.5

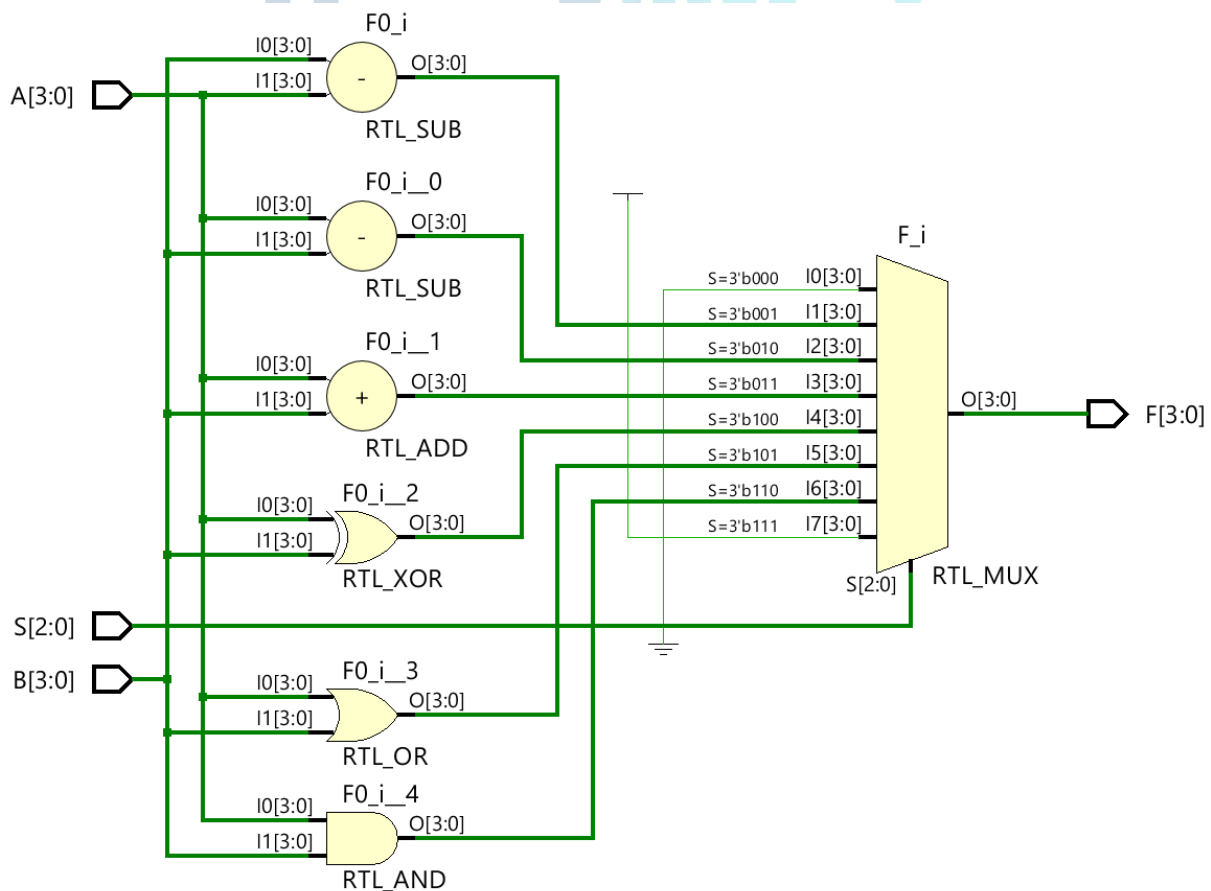


Task #1.3

```

1 module IC74381(
2     input logic [3:0] A,
3     input logic [3:0] B,
4     input logic [2:0] S,
5     output logic [3:0] F
6 );
7
8     always_comb begin
9         case (S)
10             3'b000: F = 4'b0000;
11             3'b001: F = B - A;
12             3'b010: F = A - B;
13             3'b011: F = A + B;
14             3'b100: F = A ^ B;
15             3'b101: F = A | B;
16             3'b110: F = A & B;
17             3'b111: F = 4'b1111;
18             default: F = 4'b0000;
19         endcase
20     end
21 endmodule

```



```

2 module IC74381_tb();
3
4     logic [3:0] A;
5     logic [3:0] B;
6     logic [2:0] S;
7     logic [3:0] F;
8
9     IC74381 dut (
10         .A(A),
11         .B(B),
12         .S(S),
13         .F(F)
14     );
15
16     initial begin
17
18         for (int op = 0; op < 8; op++) begin
19             for (int i = 0; i < 16; i++) begin
20                 for (int j = 0; j < 16; j++) begin
21
22                     S = op[2:0];
23                     A = i[3:0];
24                     B = j[3:0];
25
26                     #5;
27                 end
28             end
29         end
30         #50 $finish;
31     end

```



Work distribution

Section	Farah&Joud	Tala&Horiah	Rana&Hoor&Sandra
Report	✓	✓	✓
Task 1.1	✓		
Task 1.2		✓	
Task 1.3			✓